

Suricata Setup



The workflow we are going to follow

1. Check Suricata installation
- ↓
2. Identify your network interface
- ↓
3. Open suricata.yaml
- ↓
4. Configure HOME_NET
- ↓
5. Configure network interface
- ↓
6. Install/update Suricata rules
- ↓
7. Create a custom test rule
- ↓
8. Add the rule to suricata.yaml
- ↓
9. Test the configuration
- ↓
10. Start Suricata
- ↓
11. Generate traffic
- ↓
12. Check alerts

STEP 1 — Check whether Suricata is installed

Open your Kali terminal:

```
suricata --build-info
```

or:

```
suricata -V
```

You should see something similar to:

```
This is Suricata version 8.x.x
```

If the command is not found, install it:

```
sudo apt update  
sudo apt install suricata -y
```

Then verify:

```
suricata -V
```

STEP 2 — Find your network interface

Before changing `suricata.yaml`, we need to know which network interface Suricata should monitor.

Run:

```
ip addr
```

You may see something like:

```
1: lo:  
2: eth0:  
3: wlan0:
```

Or, because you're using Kali in VirtualBox, you may see something such as:

```
2: eth0@if...
```

or:

```
2: enp0s3:
```

Look for the interface that has your IP address.

For example:

```
2: eth0:  
   inet 192.168.1.15/24
```

Here:

```
eth0
```

is the interface.

Easier command

Run:

```
ip route
```

You might see:

```
default via 192.168.1.1 dev eth0
```

That tells you:

```
Network interface = eth0
```

Don't blindly use `eth0` if your system shows another interface.

STEP 3 — Back up your Suricata configuration

Before editing anything, make a backup.

Run:

```
sudo cp /etc/suricata/suricata.yaml /etc/suricata/suricata.yaml.backup
```

Verify:

```
ls -l /etc/suricata/
```

You should now have:

```
suricata.yaml  
suricata.yaml.backup
```

This is a very good habit when working with security tools.

STEP 4 — Open `suricata.yaml`

Run:

```
sudo nano /etc/suricata/suricata.yaml
```

You are now inside the Suricata configuration file.

This file is **very large**.

Don't try to understand everything inside it at once.

For our first setup, we're primarily interested in:

```
HOME_NET
af-packet
default-rule-path
rule-files
```

STEP 5 — Configure `HOME_NET`

Inside nano, press:

```
CTRL + W
```

Then search for:

```
HOME_NET
```

You'll find something similar to:

```
HOME_NET: "[192.168.0.0/16]"
```

Depending on your version, the default may already contain several private network ranges.

`HOME_NET` basically tells Suricata:

"These are the networks I consider to be my local/internal networks."

For example, if your Kali machine has:

```
192.168.1.15
```

on a:

192.168.1.0/24

network, you can configure:

```
HOME_NET: "[192.168.1.0/24]"
```

Important

Don't automatically replace the existing `HOME_NET` with that value if your lab uses several networks. The official configuration recommends that `HOME_NET` include the monitored interface's IP/network and other local networks in use.

For a simple one-network lab, however:

```
HOME_NET: "[192.168.1.0/24]"
```

is easy to understand.

STEP 6 — Configure the network interface

Now search for:

```
af-packet
```

Press:

```
CTRL + W
```

and type:

```
af-packet
```

You should find a section resembling:

```
af-packet:
- interface: eth0
```

Your interface must match the interface you discovered earlier.

For example, if:

```
ip route
```

showed:

```
default via 192.168.1.1 dev eth0
```

then:

```
af-packet:
  - interface: eth0
```

If yours is:

```
enp0s3
```

then use:

```
af-packet:
  - interface: enp0s3
```

Suricata's documentation specifically notes that the interface configured under `af-packet` needs to match the actual interface you want to monitor.

A basic configuration may look like:

```
af-packet:
  - interface: eth0
    cluster-id: 99
    cluster-type: cluster_flow
    defrag: yes
    tpacket-v3: yes
```

STEP 7 — Configure the rule path

Now search inside `suricata.yaml` for:

```
default-rule-path
```

You should find something similar to:

```
default-rule-path: /var/lib/suricata/rules
```

Then look for:

```
rule-files:
  - suricata.rules
```

This tells Suricata:

Look in `/var/lib/suricata/rules` and load `suricata.rules`.

That is the standard configuration used with `suricata-update`.

STEP 8 — Install/update the Suricata rules

Exit nano for now:

```
CTRL + X
```

Then run:

```
sudo suricata-update
```

This downloads/manages the default Emerging Threats Open ruleset and places the resulting rules in:

```
/var/lib/suricata/rules/suricata.rules
```

according to the official Suricata documentation.

Check:

```
ls -lh /var/lib/suricata/rules/
```

You should see:

```
suricata.rules
```

STEP 9 — Create your own custom rule

This is where the setup becomes much more interesting.

We're going to create:

```
local.rules
```

Create the file:

```
sudo nano /etc/suricata/rules/local.rules
```

Put this test rule inside:

```
alert icmp any any -> any any (msg:"SURICATA ICMP TEST ALERT"; sid:1000001;  
rev:1;)
```

Let's break this rule down:

```
alert
```

Generate an alert.

```
icmp
```

Monitor ICMP traffic.

```
any any
```

Any source IP and any source port.

```
->
```

Traffic direction.

```
any any
```

Any destination IP and any destination port.

Then:

```
msg:"SURICATA ICMP TEST ALERT";
```

The message that will appear when the rule triggers.

And:

```
sid:1000001;
```

The unique signature ID.

Finally:

```
rev:1;
```

Rule revision number.

Suricata rules consist of an action, header, and rule options; `alert` is one of the supported actions.

STEP 10 — Add `local.rules` to `suricata.yaml`

Open the configuration again:

```
sudo nano /etc/suricata/suricata.yaml
```

Find:

```
default-rule-path:
```

Make sure it is:

```
default-rule-path: /var/lib/suricata/rules
```

Then find:

```
rule-files:
```

You should have something like:

```
rule-files:
- suricata.rules
```

Add your custom rule:

```
rule-files:
- suricata.rules
- /etc/suricata/rules/local.rules
```

Notice that we're using the **full path** for `local.rules`.

This is supported by Suricata's configuration: rule files can be specified using either their base name or an absolute path.

STEP 11 — Save the file

Inside nano:

```
CTRL + O
```

Press:

```
ENTER
```

Then:

```
CTRL + X
```

STEP 12 — Test your Suricata configuration

This is **VERY IMPORTANT**.

Don't immediately start Suricata.

First test the configuration.

Run:

```
sudo suricata -T -c /etc/suricata/suricata.yaml
```

The important parts are:

-T

means:

Test configuration.

And:

```
-c /etc/suricata/suricata.yaml
```

means:

Use this configuration file.

You want to see something indicating that the configuration loaded successfully.

For example:

```
Configuration provided was successfully loaded.
```

and:

```
Notice: suricata: This is Suricata version ...
```

If you see an error instead, **don't start Suricata yet**.

STEP 13 — Specifically test whether your rule loads

You can also test your rule directly:

```
sudo suricata -T -c /etc/suricata/suricata.yaml
```

Look through the output for something indicating the rule was loaded.

If you want to search for your custom SID:

```
sudo suricata -T -c /etc/suricata/suricata.yaml 2>&1 | grep 1000001
```

You can also search for the alert message:

```
sudo suricata -T -c /etc/suricata/suricata.yaml 2>&1 | grep "SURICATA ICMP  
TEST ALERT"
```

STEP 14 — Check that the rule exists

Run:

```
sudo grep -n "SURICATA ICMP TEST ALERT" /etc/suricata/rules/local.rules
```

You should get:

```
1:alert icmp any any -> any any (msg:"SURICATA ICMP TEST ALERT"; sid:1000001;  
rev:1;)
```

Good.

Your rule exists.

STEP 15 — Start Suricata

Now that the configuration passes the test, you can start Suricata.

If you're using the system service:

```
sudo systemctl restart suricata
```

Then check:

```
sudo systemctl status suricata
```

You want:

Active: active (running)

The official quickstart similarly uses `systemctl restart suricata` and checks `/var/log/suricata/suricata.log` afterward.

STEP 16 — Check the Suricata log

Run:

```
sudo tail -f /var/log/suricata/suricata.log
```

You should see Suricata starting.

Press:

CTRL + C

when you're done watching.

STEP 17 — Generate traffic to trigger your rule

Now we test whether the rule actually works.

Open **another terminal window**.

Find your Kali IP:

```
ip addr
```

Suppose your IP is:

```
192.168.1.15
```

Now ping another machine on your lab network:

```
ping -c 4 192.168.1.1
```

Or, if you have another VM such as Ubuntu:

```
ping -c 4 <UBUNTU-IP>
```

Your rule is watching ICMP traffic:

```
alert icmp any any -> any any
```

Therefore, Suricata should generate an alert.

STEP 18 — Check `fast.log`

Open another terminal:

```
sudo tail -f /var/log/suricata/fast.log
```

You should eventually see something similar to:

```
[**] [1:1000001:1] SURICATA ICMP TEST ALERT [**]
```

That is the moment you know:

```
Traffic
  ↓
ICMP packet
  ↓
Suricata receives packet
  ↓
Suricata evaluates rules
  ↓
Rule SID 1000001 matches
  ↓
ALERT
```

Your first custom Suricata rule is working.

STEP 19 — Check `eve.json`

Suricata also normally records structured events in:

```
/var/log/suricata/eve.json
```

You can watch alerts with:

```
sudo tail -f /var/log/suricata/eve.json
```

For easier viewing:

```
sudo tail -f /var/log/suricata/eve.json | jq
```

If jq isn't installed:

```
sudo apt install jq -y
```

Then:

```
sudo tail -f /var/log/suricata/eve.json | jq
```

Look for:

```
"event_type": "alert"
```

and your message:

```
SURICATA ICMP TEST ALERT
```

Your complete lab

At this point your directory structure should look approximately like this:

```
/etc/suricata/  
├── suricata.yaml  
└── rules/  
    └── local.rules
```

And your main rule database:

```
/var/lib/suricata/rules/  
└── suricata.rules
```

Your logs:

```
/var/log/suricata/  
├── suricata.log  
├── fast.log  
└── eve.json
```

The commands you'll use repeatedly

Once everything is configured, your normal workflow becomes:

1. Check interface

```
ip addr
```

2. Update rules

```
sudo suricata-update
```

3. Test configuration

```
sudo suricata -T -c /etc/suricata/suricata.yaml
```

4. Restart Suricata

```
sudo systemctl restart suricata
```

5. Check status

```
sudo systemctl status suricata
```

6. Watch alerts

```
sudo tail -f /var/log/suricata/fast.log
```

7. Watch structured events

```
sudo tail -f /var/log/suricata/eve.json | jq
```

One important distinction

There are **two different things** here:

suricata.yaml

This is the **configuration file**.

It tells Suricata things such as:

What network should I monitor?

Which interface should I use?

Where are my rules?

Where should I write logs?

What protocols/features are enabled?

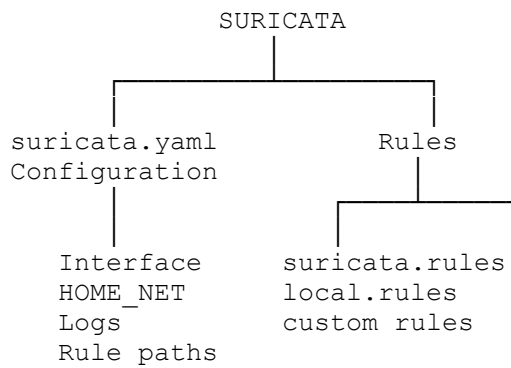
local.rules

This is your **actual detection rule file**.

For example:

```
alert icmp any any -> any any (msg:"SURICATA ICMP TEST ALERT"; sid:1000001;  
rev:1;)
```

So think of it this way:



Suricata's official documentation recommends `suricata-update` for managing the main ruleset, while custom rules can be added separately and referenced from `suricata.yaml`.